PACE

Technical Design Document

Field collections, intelligently prioritised.

A production-style, offline-first collections operating system for an RBI-registered NBFC serving small manufacturers and traders in Indore.

Repository	https://github.com/Aryan-coder06/Pace-Field-collections-intelligently-prioritised..git
Live application	Not deployed from this workspace. Hosting credentials and explicit authorization are required before publishing a synthetic demo.
Build status	Stage 4 hardening completed locally: API tests, browser journeys, lint, typecheck and production build verified.
Data policy	Synthetic borrowers, synthetic history, synthetic GPS examples and mock payment/reminder workflows only.

Prepared for hackathon review - 8 November 2026

1. Executive summary

Pace replaces the lender's daily overdue Excel and WhatsApp hand-off with a branch-scoped operating system. Managers validate an overdue file, retain dated snapshots, make explainable assignments and inspect operational performance. Field officers receive a concise risk-aware queue, record outcomes and promises, and preserve eligible work through poor connectivity.

The build favors correctness over surface complexity: monetary values are decimal-backed; snapshots are append-only; assignments and promises are versioned; consequential actions create audit records; and AI recommendations remain advisory and reversible.

Target client outcomes are reduced roll rates, same-day visit logging and removal of weekly Excel reconstruction. These are operational targets, not measured production claims.

Core delivered

Capability	Implemented outcome
Collections lifecycle	Validated CSV -> dated snapshot -> deterministic assignment -> outcome -> promise lifecycle -> analytics.
Offline resilience	User/branch scoped Dexie cache, durable outbox, per-event acknowledgement and visible conflicts.
AI decision support	Regularized logistic demonstration, grounded reasons, holdout metrics, back-test and audited rank override.
Compliance	Contact window, object-level RBAC, CSRF, audit ledger and synthetic-only data.

2. Architecture

Pace is a modular monolith. At approximately 55 officers and 4,800 loans, a single deployable FastAPI application with separated domains is more reliable and easier to operate than microservices.

Next.js PWA App Router, TanStack Query, Dexie outbox, shell cache	FastAPI Auth, imports, assignments, collections, promises, analytics, sync, AI	PostgreSQL Snapshots, versioned state, receipts and audit events

Key boundaries

Layer	Responsibility	Safety property
Web	Manager workspace and mobile officer workflow.	Never labels a device save as server-synced.
API	Authorization, validation, transactions and policy.	Server derives actor and branch scope.
Database	Projections plus append-only history.	UUID receipts, versions and import guards.
Offline client	Minimal cache and explicit outbox lifecycle.	No tokens in IndexedDB; scoped cache.

Why not microservices?

Import, assignment, outcomes, promises and analytics require local transactions and common audit semantics. Domain seams remain clean for a future extraction, without introducing distributed

operational failure modes now.

3. Data model and auditability

Loan identity is separate from daily condition. LoanSnapshot is never overwritten by import processing; the loan/snapshot-date pair is unique. That preserves the history needed for roll-rate calculation and model training.

Branch / User role and branch scope	Borrower / Loan stable synthetic identity	LoanSnapshot dated DPD and balances	Assignment versioned ownership	Outcome / Promise field evidence and lifecycle	Audit / Receipt immutable trace

Entity	Purpose	Integrity control
LoanSnapshot	Dated outstanding, overdue and DPD.	Unique loan + date; same-day conflicts require review.
Assignment	Officer ownership for a business day.	Active guard, branch/pincode checks and version.
CollectionOutcome	Reported field result and location state.	Immutable device event; PAID does not reduce outstanding.
PromiseTransition	Commitment status history.	One open promise; KEPT only through explicit mock verification.
SyncEventReceipt	Server acknowledgement for client UUID.	Unique event prevents duplicate replay.
AuditEvent	Actor, branch, entity, action and metadata.	Append-only operational trace.

4. Main workflows

Daily import and assignment

Upload UTF-8 and size limits	Preview row errors and duplicates	Confirm transactional snapshots	Assign pincode + normalized cost	Review exceptions and overrides

Assignment chooses the lowest normalized sum of account count, outstanding exposure and 30+ DPD accounts among active in-branch officers who cover the pincode. Stable user-ID tie-breaking is deterministic. Unsupported pincodes remain unassigned.

Officer outcome and promise

Open account authorized assignment	Select four outcome types	Persist transaction + client UUID	Promise same transaction	Refresh queue, tracker, activity

DISPUTE requires explanation. PROMISE_TO_PAY requires positive amount and valid due date. PAID is only reported field evidence pending an authoritative settlement integration.

Offline synchronization

Cache minimum authorized queue	Outbox persist before UI success	Health API reachability	Sync batch UUID replay	Resolve acknowledged/conflict/rejected

A persisted SYNCING event is replayed in foreground recovery. Server idempotency makes replay safe. A stale assignment becomes an ASSIGNMENT_REASSIGNED conflict rather than a force-applied outcome.

5. AI architecture and governance

The model is an explainable synthetic demonstration, not a production-calibrated financial model. It estimates roll-forward risk from stored snapshot transitions and informs an operational score. A manager's effective priority may change ordering while preserving original model evidence.

Metric	Measured synthetic holdout
Model	pace-logit-v2-1-20261107172635133161
Observations / positives	152 / 14
Temporal split	122 training / 30 validation
ROC-AUC	0.5972
Accuracy / precision / recall	0.7333 / 0.3333 / 0.3333
Brier score	0.1956

Explanation and oversight

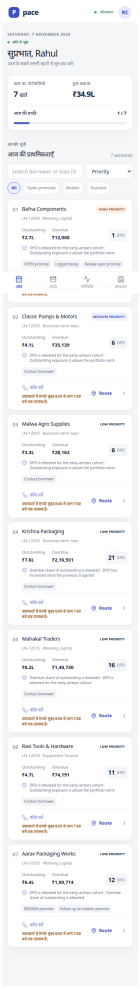
Concept	Meaning
Learned roll-risk estimate	Probability-oriented output of the logistic model on available synthetic features.
Operational priority score	Risk-aware score used to make officer work scannable.
Effective manager priority	Versioned rank override with retained rationale; never overwrites model result.
Reasons	Three largest positive standardized feature contributions expressed in controlled business language.

The synthetic cohort is small and validation performance is modest. Pace labels the output advisory, provides deterministic fallback and makes no claim of collection uplift or autonomous decision authority.

6. UI/UX design

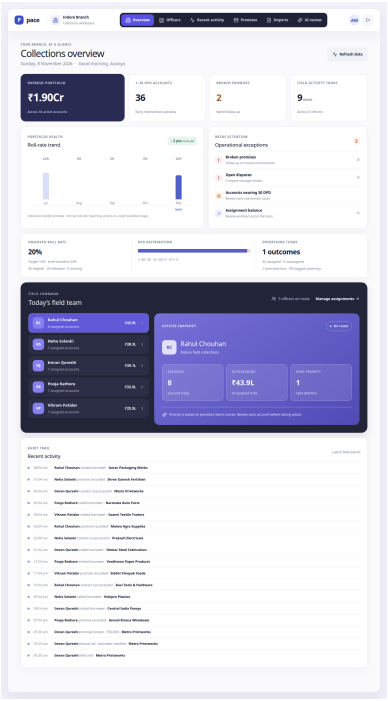
The visual system uses a quiet near-white canvas, deep ink typography, restrained indigo actions, tabular financial numbers and thin separators. It avoids generic dashboard noise so operational signals remain legible outdoors and on low-end Android devices.

Officer Today at 375px

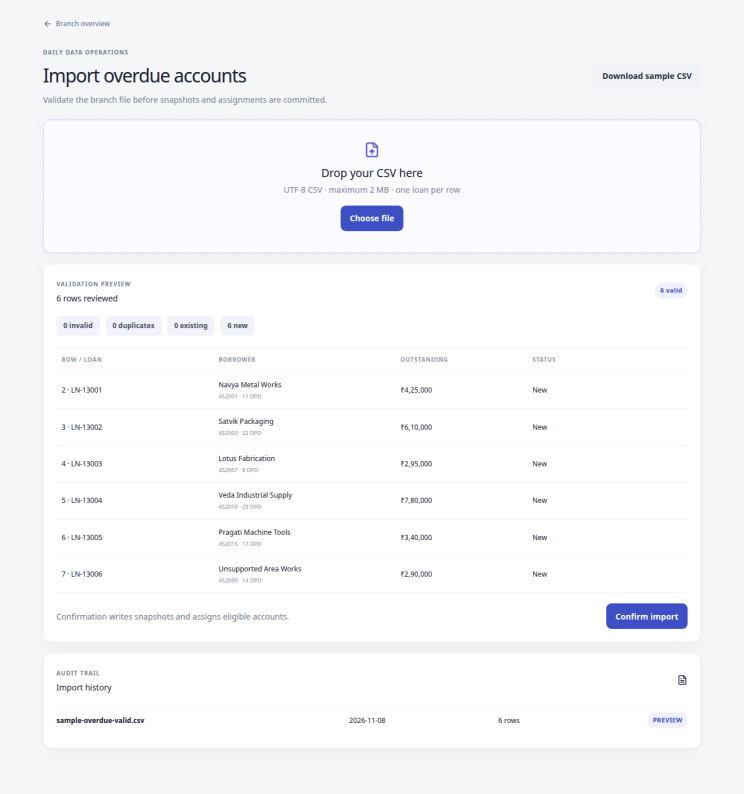


The compact mobile queue keeps borrower, outstanding, DPD, reasons, promise context and the next action readable in seconds. Bottom navigation stays thumb reachable.

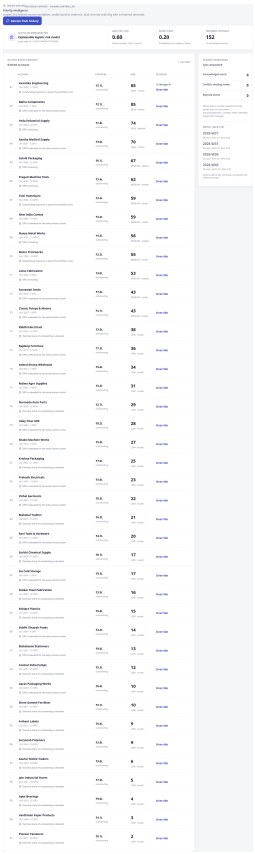
Manager overview



Import validation console


Branch overview
DAILY DATA OPERATIONS
Import overdue accounts
Download sample CSV
Validate the branch file before snapshots and assignments are committed.
Drop your CSV here
UTF-8 CSV · maximum 2 MB · one loan per row
Choose file
VALIDATION PREVIEW
6 rows reviewed
6 valid
0 invalid
0 duplicates
0 existing
6 new
ROW / LOAN
BORROWER
OUTSTANDING
STATUS
2 · LN-13001
Navya Metal Works
452001 · 11 DPD
₹4,25,000
New
3 · LN-13002
Satvik Packaging
452003 · 22 DPD
₹6,10,000
New
4 · LN-13003
Lotus Fabrication
452007 · 8 DPD
₹2,95,000
New
5 · LN-13004
Veda Industrial Supply
452010 · 29 DPD
₹7,80,000
New
6 · LN-13005
Pragati Machine Tools
452015 · 17 DPD
₹3,40,000
New
7 · LN-13006
Unsupported Area Works
452099 · 14 DPD
₹2,90,000
New
Confirmation writes snapshots and assigns eligible accounts.
Confirm import
AUDIT TRAIL
Import history
sample-overdue-valid.csv
2026-11-08
6 rows
PREVIEW

AI review and human control



7. Security, privacy and compliance posture

Control	Implemented behavior
Authentication	Argon2 password hashing, HttpOnly access/refresh cookies, refresh rotation and logout revocation.
Mutation protection	Double-submit CSRF validation for cookie-authenticated mutations.
Authorization	Backend role checks plus assignment/branch object scope; frontend routing is not the security boundary.
Contact policy	Server-authoritative 08:00 inclusive to 19:00 exclusive Asia/Kolkata enforcement; blocked attempts audited.
Offline privacy	No token in IndexedDB; minimal user/branch scoped cache; fresh offline restart cannot bypass authentication.
GPS	Browser location status only; absent/low accuracy state is retained and never invented.
Data/messaging	Synthetic records only; reminders are approved templates and simulated delivery.

Pace implements safeguards but does not claim RBI, DPDP or production compliance certification. Shared-device policy, managed-device controls, encryption-at-rest and incident operations require organization-level deployment work.

8. Testing and validation

All results below were executed locally during final hardening. Browser tests used a disposable PostgreSQL database, applied all migrations, seeded synthetic records and made real FastAPI requests.

Check	Result
Backend pytest	24 passed; one upstream Starlette TestClient deprecation warning.
Backend Ruff	Passed.
Web ESLint / TypeScript	Passed.
Playwright	6 passed in 35.3 seconds.
Browser coverage	360x800, 375x812, 430x932, 768x1024, 1280x800 and 1440x900; no horizontal document overflow.
Migration / seed	Clean PostgreSQL migration through 0005_offline_ai and repeatable synthetic seed succeeded.

Browser journeys

Journey	Verified result
Manager	CSV preview/confirmation, assignment override, audit visibility and AI rank override.
Officer	Authorized loan detail, promise creation, immediate tracker update and GPS capture path.
Offline	IndexedDB persistence, reconnection replay and exactly-once server outcome.
Conflict	Manager reassignment produced retained audited conflict without former-officer outcome.
Reminder	English/Hindi preview; delivery disabled outside contact window in this final run.

9. Trade-offs and roadmap

Not a production capability	Reasoned decision
Bank settlement	PAID remains a field report until verified reconciliation is integrated.
Real SMS/WhatsApp	Simulated templates avoid uncontrolled borrower communication and provider-compliance work.
Universal Background Sync	Foreground replay is the cross-browser contract.
Production ML claims	Synthetic data demonstrates pipeline mechanics, not expected uplift.
Microservices/Kubernetes	Unnecessary operational surface at the current scale.

Next: LOS and payment integration, privacy/security review, controlled provider messaging, real temporal model monitoring and a measured branch pilot.

10. Assumptions

Client facts are distinguished from engineering assumptions in docs/ASSUMPTIONS.md.

Range	Summary
001-005	Officer/branch scope, business timezone, demo accounts and seeded priority framing.
006-010	CSV contract, idempotency, pincode fallback, assignment cost and override policy.
011-020	Promise lifecycle, payment meaning, contact/GPS behavior, roll-rate coverage and synthetic data.
021-026	Offline authentication, cache scope, outbox retention, version conflicts and foreground sync.
027-031	Synthetic model, holdout, grounded explanations, human override and controlled templates.
032-035	Process-local rate limiting, SYNCING replay, Hindi boundary and deployment authorization.

No unknown production URL, benchmark or compliance claim has been invented for this submission.